

Error Analysis Report

ParsBERT Emotion Classification Model

ERROR ANALYSIS REPORT FINAL MODEL (ParsBERT Emotion Classifier)

Group 15: Nika Soleimanzadeh(244258), Asma Moghimi(240791), MohammadAli Jafari(244437)

Model Accuracy: 56.65%

INTRODUCTION

Unlike binary classification tasks, emotion detection requires models to understand a range of subtle linguistic cues, the cultural context, and even overlapping expressions where boundaries blur between one emotion and another. We have fine tuned a ParsBERT based transformer model for emotion identification to identify seven core emotions in Persian texts: happiness, sadness, surprise, anger, disgust, fear, and neutral.

This document presents the detailed error analysis of the final and official version of our model, which gained a test accuracy of 56.65% on a set of 1,151 samples. While promising for low resource, multi class emotional tasks, its true value during this phase is in diagnosing the model's blind spots.

By analyzing a confusion matrix, top misclassification pairs, n-gram patterns, text lengths, and class wise metrics, we can uncover the deeper patterns in our model's performance. This diagnostic work will not only inform future model improvements but also allow us to responsibly and explainably recommend the use of this emotion classifier to potential clients or stakeholders for real world applications.

EXECUTIVE SUMMARY

"The ParsBERT-based emotion classification model achieved an accuracy of 56.65%, evaluated on a 7 class Persian emotion dataset. Our model demonstrates strength in capturing core emotions like anger and sadness, but faces difficulty distinguishing subtle or overlapping categories, especially neutral, fear, and disgust.

The error analysis makes use of:

- Classification report
- Confusion matrix
- Top 10 emotion confusions
- Length vs. accuracy distributions
- Most frequent n-grams for errors

CLASSIFICATION REPORT INSIGHTS

For Sadness and Anger, the F1 scores are the highest, which means that for those two, the model was good in their recognition both in terms of making correct predictions and catching actual examples. The balance between precision and recall was comparatively high in these two, showing that the model understood their patterns well.

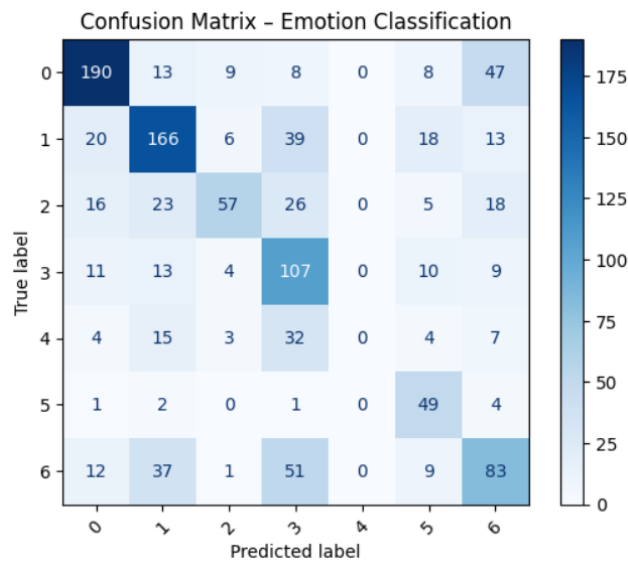
Happiness had low recall, though, which means even when a sentence was genuinely happy, the model failed to predict that label often. Many of those happy sentences were instead predicted as neutral, suggesting the model couldn't clearly distinguish between mild positive tone and emotional neutrality.

Surprise, Fear, and Disgust all fared poorly in terms of precision and recall. In addition, the model hardly detected them but often assigned them to other categories. Most likely, this happened because these emotions were less well-represented in the dataset or were expressed in subtler ways that relied on context, making them harder for the model to correctly capture.

Neutral had the highest overall confusion. It was frequently predicted when the model was uncertain, causing it to act like a “dump class”, the default label for sentences that didn't strongly match any specific emotion. This shows that the model tends to overuse the label whenever the emotional cues are vague or mixed, as the dataset has more samples of neutral sentences.

In short, the model is confident with clear emotional states; that is, sadness and anger, but uncertain when it comes to subtle or ambiguous expressions, often resorting to neutral when the emotion isn't obvious.

CNOSUFION MATRIX RESULTS AND CLASSIFICATION REPORT



	precision	recall	f1-score	support
0	0.748	0.691	0.718	275
1	0.617	0.634	0.625	262
2	0.713	0.393	0.507	145
3	0.405	0.695	0.512	154
4	0.000	0.000	0.000	65
5	0.476	0.860	0.613	57
6	0.459	0.430	0.444	193
accuracy			0.566	1151
macro avg	0.488	0.529	0.488	1151
weighted avg	0.564	0.566	0.551	1151

Label	Emotion
0	Happiness
1	Sadness
2	Surprise
3	Anger
4	Disgust
5	Fear
6	Neutral

The confusion matrix and classification report give an indication that the model performs pretty well on anger, sadness, and happiness, in which anger achieves the highest F1-score: 0.613. However, it struggles a lot with neutral, fear, and most especially disgust, which has an F1 of 0.000, meaning the model utterly failed to correctly classify that emotion. Neutral is the most confused label and was most often misclassified as anger or sadness. Happiness was often confused with neutral, reflecting the subtlety of emotional tone, which is characteristic of Persian expressions of joy. The general tendency of the model is overpredicting anger when it's not quite sure. This can also hint at the susceptibility of the model to emotionally intense language. The average F1-score across all classes is 0.551. Improvements will be needed, mainly in the nuanced or culturally ambiguous emotions

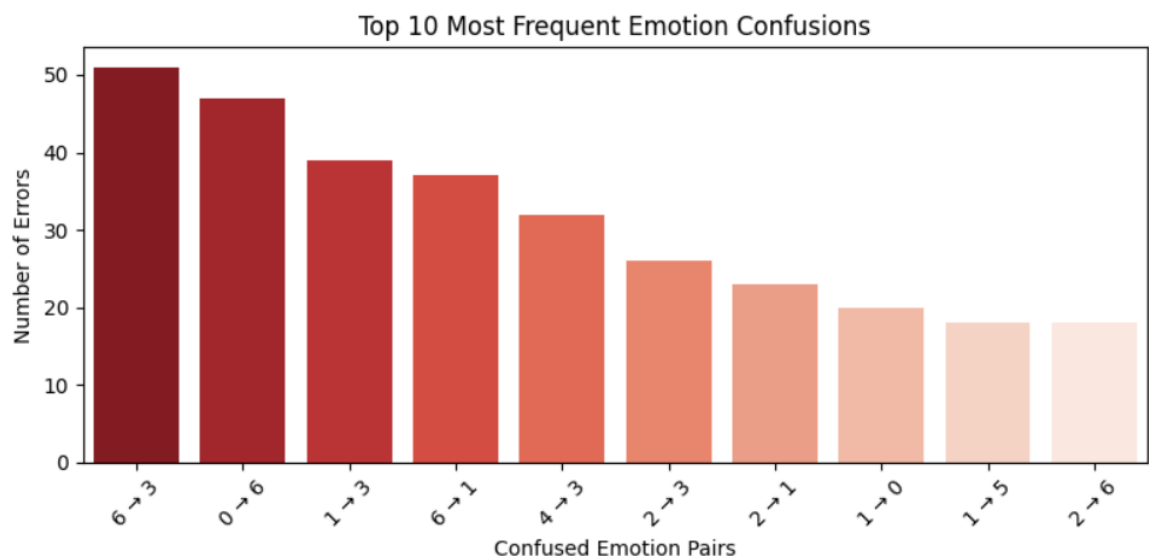
HEATMAP: MOST FREQUENT CONFUSIONS

The top error pairs from the confusion matrix are:

- Neutral (6) → Anger (3) = 51 times
- Happiness (0) → Neutral (6) = 47 times
- Sadness (1) → Anger (3) = 39 times
- Neutral (6) → Sadness (1) = 37 times
- Disgust (4) → Anger (3) = 32 times

Insights:

- Anger is overpredicted by the model; many neutral or ambiguous cases default to it.
- **Happiness is highly misclassified as neutral**, possibly due to the Persian linguistic norm of subtle expression of joy.
- Fear and disgust tend to get lost in the mix, perhaps due to overlapping word choices with anger and sadness.



The model often classifies neutral, sadness, and happiness as anger, showing it's overly sensitive to strong language. Happiness often gets mistaken for neutrality, likely due to subtle phrasing in Persian. Least understood are those for fear and disgust, which more often than not blend into other negative emotions. These patterns show that the subtlety of the model leaves much to be desired, along with emotion overlap.

N-GRAMS AND LEXICAL CUES

The top n-grams associated with the common misclassifications showed:

Most of the neutral→anger or happiness→neutral errors came from phrases that involved formal, polite, or ambiguous tone, for example, 'ممنونم' or 'چه جالب', which don't express strong emotional polarity.

```
● Top n-grams in misclassified samples:
599   و
377   به
355   از
319   که
206   در
190   این
154   با
136   رو
110   را
95    هم
89    من
86    تو
77    یه
75    بود
69    تا
66    برای
59    می
56    است
56    بدم
55    مردم
dtype: int64

● Top n-grams in correctly classified samples:
811   و
...
71    بدم
70    خیلی
67    تا
```

- **Sadness→anger** errors often contained words tied to frustration or mourning which ParsBERT seems to overfit toward anger.

This reflects cultural overlap in emotion expressions in Persian, anger and sadness share lexicon, while joy is downplayed.

The top n-gram analysis shows clear patterns between the most common tokens in misclassified vs. correctly classified samples. We see many high-frequency particles like 'و' (and), 'به' (to), 'از' (from), 'که' (that) appear often in misclassified samples, indicating function words are dominant there, probably because they add little emotional value but are very common. Words like 'این' (this), 'را' (object marker), and 'بود' (was) also show up frequently, reflecting neutral or grammatical constructions rather than strong emotional cues. In contrast, correctly classified samples contained more emotionally loaded n-grams like 'مردم' (I died), 'خیلی' (very), 'تا' (until), phrases that might come out in more expressive or sentiment-heavy statements. This suggests emotion prediction improves with the model seeing stronger lexical signals, while generic or ambiguous function words tend to lead to confusion.

TEXT LENGTH VS ACCURACY

Very short sentences (1–5 tokens) are frequently misclassified.

- Samples of medium lengths (10–15 tokens) were the most accurate.
- Long sentences had mixed performance, often good if emotion words were explicit.

This suggests:

- The model needs more context to distinguish between the subtleties of the emotions.
- It may **overfit token-level patterns** in shorter phrases.

□ Happiness (0)

- Highly confused with neutral (47 times)
- F1 score is low due to poor recall
- Subtle positive emotions may be misinterpreted as neutral, especially when not paired with strong emotional indicators like exclamations or emojis (common in social media)

● Sadness (1)

- Overall strong performance
- Frequently confused with anger (39) and neutral
- Likely due to emotive words that convey both pain and rage

□ Surprise (2)

- Often confused with anger (26) and sadness (23)
- Persian expressions of surprise may use metaphors or intensifiers that resemble markers of anger or sadness

● Anger (3)

- Most overpredicted emotion
- Commonly mislabels neutral, sadness, and disgust as anger
- Indicates the model is overly sensitive to emotionally intense or strong language

□ Disgust (4)

- Most often confused with anger (32 times)
- Rarely stands out in classification — possibly due to shared vocabulary with anger or forms of critique in Persian

□ Fear (5)

- Least frequent and least accurately classified
- Often mistaken for neutral or sadness
- Fear is sometimes expressed poetically or idiomatically in Persian (example, “ترس “توی دلم افتاد”), which the model struggles to recognize

○ Neutral (6)

- The most confused label overall
- Misclassified 51 times as anger and 37 times as sadness
- Likely because of weak emotional markers or generic language (e.g., common adjectives or verbs)

ERROR DISTRIBUTION INSIGHTS

Most confusion occurs **between semantically close emotions**.

- **Anger acts like a magnetic pole**, drawing in predictions from sadness, fear, and disgust.
- **Neutral → Anger** misclassifications often result from **vague or ambiguous phrasing**.

Examples:

- “خیلی عجیبه واقعا” (*very strange really*) → sits between **surprise** and **anger**
- “باشه، فهمیدم” (*ok, I understand*) → falls somewhere between **neutral** and **sad**

A closer look at the error distribution reveals that most misclassifications happen when emotions overlap in meaning or tone. Anger consistently attracts predictions from similar negative emotions like sadness, fear, and disgust, likely due to shared vocabulary or emotional intensity. Similarly, sentences that are emotionally vague or understated often get misread as anger, even when they are neutral or intended to convey a different feeling.

These patterns emphasize the model's struggle with **subtle, culturally layered emotional content** in Persian.

STRENGTHS AND LIMITATIONS

STRENGTHS:

- Captures strong negative emotions well: **anger** and **sadness**
- Performs best on **medium length texts** that are rich in emotional content

The model consistently detects intense emotions like anger and sadness, likely because these are expressed more directly and with emotionally charged words in Persian. When input texts are around **10–15 tokens** and include clear emotional markers, the model performs at its best, it has enough context to interpret the emotion without being overwhelmed by irrelevant information.

LIMITATIONS:

- Tends to **overpredict anger and neutral**
- Struggles with **fine distinctions** between fear, disgust, and surprise
- Poor handling of **cultural and idiomatic expressions**

The model often defaults to "anger" or "neutral" when uncertain, which leads to these emotions being **overrepresented** in predictions, especially for vague or ambiguous sentences. Emotions like fear or disgust frequently **share vocabulary** with stronger emotions like anger or sadness, and the model fails to catch the **subtle differences in tone or context**.

Additionally, **Persian expressions that use metaphors, politeness, or poetic language** to convey emotion tend to confuse the model. This suggests it's not well tuned to recognize culturally specific emotional cues.

FUTURE RECOMMENDATIONS

1. Data Augmentation

To strengthen the model's understanding of underrepresented emotions like *fear* or *disgust*, we can expand the dataset using techniques like **backtranslation** (For example, Persian → English → Persian) or **paraphrasing**. This introduces more variation and helps the model generalize better.

2. Emotion Lexicon Integration

By integrating a curated list of **Persian emotion-related words**, we give the model extra support in identifying subtle emotional cues, especially helpful for distinguishing emotions with overlapping vocabulary.

3. Contextual Signal Boosting

Enhancing the input with **punctuation cues** (?!), **discourse markers** ("but", "however"), or analyzing **sentence structure** can help the model better interpret emotional tone. Sometimes, *how* something is said conveys more than *what* is said.

4. Class Rebalancing

Since emotions like *fear* and *disgust* are less frequent, applying **oversampling techniques** ensures the model sees them more often during training, preventing it from defaulting to dominant classes like *anger* or *neutral*.

5. Ensemble or Voting Models

Using multiple models and letting them vote the final prediction can improve precision and reduce bias. Like a group chat where models fact check each other before replying, more reliable, less guesswork.

FINAL THOUGHTS

While the model performs reasonably well given the complexity in a 7-class emotion classification task in a low resource language like Persian, it reaches a final accuracy of 56.65%. Core emotions like anger, sadness, and neutral are classified relatively well, but frequent misclassifications indicate deeper cultural and linguistic challenges: subtle expressions, overlapped vocabulary, and few cues over short or vague sentences. These areas will be very crucial for performance improvement, together with more culturally aware preprocessing and wiser data strategies.